UNIT 8 DISASTER MANAGEMENT: FINANCIAL ARRANGEMENTS AND MANAGEMENT*

Structure

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8.0 OBJECTIVES

After studying this Unit, you should be able to:

- Describe the concept and mechanisms of financial arrangements for disaster risk reduction and management in India;
- Explain the recommendations of the Finance Commission on disaster management;
- Discuss the relevant policies of disaster risk financing and management in India;
- Examine the issues and challenges in disaster risk financing; and
- Highlight the way forward for effective financial arrangements and disaster management.

8.1 INTRODUCTION

Disasters, natural and human-induced, have profound economic impacts that can destabilise the economy, decelerate economic development, and increase poverty. The increasing frequency and intensity of disasters due to climate change and other factors underscore the urgent need for robust financial strategies to manage

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these risks. In this context, investing in Disaster Risk Reduction (DRR) is not only a cost-effective approach to safeguarding lives and livelihoods but also a crucial element in sustaining economic development. Financial management plays a greater role in the entire gamut extending from disaster risk management to preparedness, response, recovery, etc. Financial management for disasters refers to the strategic planning, organising, directing, and controlling of financial activities at various levels of an institutional framework. It involves managing the organisational finances to achieve financial goals including disaster risk financing, risk insurance, financing on prevention, preparedness and mitigation; post-disaster financial arrangements, etc.

Financial strategies for managing the economic impact of disasters comprise a broad range of interventions which include investments in preparedness, mitigation, response, and recovery efforts, as well as innovative funding mechanisms that ensure adequate resources are available at each stage of disaster management. The economics of disaster management highlights the importance of allocating resources efficiently to minimise losses and ensure rapid recovery.

In India, the institutional mechanisms for disaster financing play a greater role in supporting Disaster Risk Reduction (DRR) and Disaster Risk Management (DRM) initiatives at all levels. Thus, understanding the institutional mechanism of financing in different phases of disaster management is crucial for optimising financial resources and enhancing resilience.

Mainstreaming DRR into developmental interventions requires a strategic, contextual-based investment that addresses both current and future risks. Effective coordination, among national, state, district and sub-districts, as well as collaboration with donor agencies, is essential for ensuring that financial resources are used efficiently and effectively. By prioritising "ex-ante" investments, which focus on reducing risks before a disaster occurs, alongside "ex-post" investments for recovery and rebuilding, disaster impacts can be better managed; and developmental gains can be protected. This will not only strengthen economic development and community resilience but also contribute to achieving Sustainable Development Goals (SDG) by reducing vulnerabilities and building a safer and more secure future. In this Unit, we will discuss the mechanisms of Financial Arrangements for Disaster Risk Reduction and Management in India, major recommendations of the Finance Commissions on Disaster Management, relevant policies of Disaster Risk Financing and Management in India, issues and challenges in Disaster Risk Financing, and the way forward for effective financial arrangements and disaster management.

8.2 DISASTER MANAGEMENT: FINANCIAL MECHANISM

The financial mechanism of Disaster Management in India is largely based on the provisions of the Disaster Management Act, 2005 (The DM Act, 2005). In this context, Chapter IX of the Act deals with Finance, Accounts and Audits, wherein sections 46 to 50 have prescribed the following provisions:

Section 46 – National Disaster Response Fund,

Section 47 – National Disaster Mitigation Fund,

Section 48 – Establishment of funds by the State Government,

Section 49 – Allocation of funds by Ministries and Departments, and

Section 50 – Emergency Procurement and Accounting.

It has been observed that the past practices of Disaster Financing in India were centred on response, relief and rehabilitation by way of the institutional framework of the Calamity Relief Fund (CRF), and National Calamity Contingency Fund (NCCF). Later on, the NCCF was renamed the National Disaster Response Fund (NDRF) under the provision of The DM Act, 2005.

Financial Arrangements at the National Level

Disaster Management being the state subject, the prime responsibility for providing response support including rescue, relief rehabilitation etc. lies with the State Government. However,

in case a disaster of severe magnitude occurs in any State/UT and the available fund under the *State Disaster Response Fund (SDRF)* proves insufficient for disaster response, then the Central Government may provide assistance by using the *NDRF* by following the laid down procedures. As per this procedure, the Government of disaster-affected State/UT is required to submit a "*memorandum"* wherein all details of sector-specific damages are furnished to the Ministry of Home Affairs, Government of India along with the required fund. Afterwards, an Inter-Ministerial Central Team visits the affected State for assessment and to find the scope of fund disbursement as per the norms of the SDRF and NDRF.

The Sub-Committee of the National Executive Committee (NEC) examines the State's request along with the Assessment Report of the "Inter-Ministerial Central Team" as per Section-6 of the Disaster Management Act, 2005, and accordingly recommends the amount that can be funded from the NDRF. Based on these recommendations, the High-Level Committee (HLC) approves the disbursement of relief funds from the NDRF.

The Disaster Management Division, Ministry of Home Affairs, GoI, oversees the utilisation of funds and ensures smooth compliance at the State level.

Financial Arrangements at the State Level

Under Section 48 (1)a of the Disaster Management Act, 2005, it is provisioned that the *State SDRF* may be utilised by the States/UTs solely for providing immediate relief to victims of various disasters, including *cyclones, droughts, earthquakes, fires, floods, tsunamis, hailstorms, landslides, avalanches, cloudbursts, pest attacks, frost,* and *cold waves*.

In addition to the SDRF being utilised by states for emergency response and relief efforts, they can also adjust a portion of these expenses against funds released from the NDRF. This adjustment applies to costs incurred in the fiscal year preceding the release of the NDRF.

It is also significant to note that if a state experiences multiple severe disasters within the same year, assistance from the NDRF will not be reduced, ensuring continued support during crises. In case of state-specific disasters, which are not included in the "officially notified list for SDRF and NDRF" assistance, states can use up to 10 per cent of their annual SDRF allocation to address localised or state-specific disasters.

There are few essential activities, which are covered under the provisions of the NDRF and SDRF. These include Gratuitous Relief; Search and Rescue Operations, based on actual costs incurred; Relief to Affected Community; Airdropping Essential Supplies; Emergency Drinking Water Supply; Debris Management; Immediate Support for Agriculture, Animal Husbandry, Fisheries, Handicrafts, and Artisans; Immediate Repair and Restoration of Damaged Infrastructure; and Capacity Development.

The default period for assistance is set according to the established norms. However, the State Executive Committee (SEC) may extend this period based on ground assessments, provided that the additional expenditure does not exceed 25 per cent of the SDRF allocation for that year. The SEC is responsible for managing contributions from relevant State departments, administering the SDRF, and investing funds following guidelines approved by the GoI. Typically, a Needs Assessment is carried out by the SEC. Significantly, Capacity Development expenses are provisioned to be covered by the SDRF, not the NDRF, with a limit of 10 per cent of the SDRF. This may include managing Emergency Operation Centres (EOCs) within the State; training and capacity-building for Stakeholders; supporting state disaster management institutions; formulation of Disaster Management Plans; and strengthening the State Disaster Management Authorities (SDMAs) and District Disaster Management Authorities (DDMAs).

For disasters requiring central support beyond the SDRF, the Ministry of Home Affairs processes requests from the state governments for assistance from the Government of India. The Ministry of Finance is responsible for making the necessary budgetary provisions for relief funds aimed at strengthening the response mechanisms, enhancing disaster management institutions, and supporting the capacity development of stakeholders, as well as disaster risk reduction efforts.

Now, we will focus on the financial allocation to the centre and states.

8.3 FINANCIAL ALLOCATION TO THE CENTRE AND STATES: RECOMMENDATIONS OF THE FINANCE COMMISSIONS

The financial strategy for managing the economic impact of disasters necessitates a comprehensive, systematic, and coordinated approach. This approach is essential not only for managing disaster response but also for promoting disaster resilience. The Finance Commission (FC) of India plays a pivotal role in distributing financial resources between the Central government and the States/UTs. Over the years, recommendations from the Tenth to the Fifteenth Finance Commission have consistently emphasised the need for adequate and balanced funding. These recommendations are aimed at strengthening the financial and institutional frameworks for disaster management in India, ensuring that both the Union Government and the States/UTs are better prepared to tackle the challenges posed by disaster The specific recommendations made by the Finance Commissions (Tenth to Fifteenth FC) regarding Disaster Management are as follows:

i) The Tenth Finance Commission

The Tenth Finance Commission of India was constituted **under the Chairmanship of Mr. K.C. Pant** for the period 1995-2000, which made several key recommendations regarding disaster management funding. Recognising the increasing frequency and severity of natural disasters in India, the Commission recommended the establishment of a dedicated fund known as the *Calamity Relief Fund (CRF)*. This fund was intended to be used exclusively for immediate relief and rehabilitation efforts following natural disasters. The recommendations laid the foundation for a more structured and systematic approach to disaster management funding in India, ensuring that states had the necessary resources to respond promptly during the emergency.

Major Recommendations of the Commission are as follows:

- The *Calamity Relief Fund (CRF)* as introduced in the Ninth FC was recommended to be continued with modifications. The prescribed annual contribution recommended for the Centre and States/UTs was in the ratio of 75:25 respectively.

- The amount worked out for all the states for the period was 6304.27 crores, out of which the Centre contributed 4728.19 crores (75%) and states/UTs 1576.08 crores.
- The CRF was designed to enable each State/UT to manage disasters on its own and provide calamity relief by drawing upon resources available with a fund allocated for each State.
- The Tenth FC also specifically described the *"Calamity of rarest Severity"* and recommended that once a calamity is deemed to be of rarest severity, it needed to be dealt with as the "National Calamity" with required assistance and support from the Centre and supporting States/UTs beyond the general provisions under the CRF scheme.
- To effectively deal with the calamity of rarest severity, the FC recommended a landmark provision of extending solidarity and support both in terms of financial and resource support from all states/UTs to the affected state(s). The Commission proposed that in addition to CRF, the specific fund *"National Fund for Calamity Relief (NFCR)"* may be created and subscribed by the Centre and all the States/UTs in the ratio of 75:25 respectively. The total recommended size of the fund was ₹ 700 crores (built over a period 1995-2000). An initial corpus of ₹ 200 crores was created wherein contributions of ₹ 150 crores and 50 crores were made by the Centre and States/UTs respectively. In addition, the annual contribution was also fixed between the Centre and States in the ratio of 75 crores and 25 crores respectively.

Highlights of NFCR

- It is operated by the Ministry of Agriculture, Government of India with the representation of the Centre and States/UTs.
- It is managed by the National Calamity Relief Committee (NCRC).
- It is Headed by the Union Agriculture Minister and represented by the Deputy Chairman, Planning Commission, Two Union Ministers, and five Chief Ministers to be nominated by the Prime Minister annually on a rotation basis.
- The fund is to be maintained outside the Public Account of the Government of India.

ii) The Eleventh Finance Commission

This Finance Commission (**2000-2005)** under the **Chairmanship of Prof. A.M Khusro** recommended continuing the scheme of the Calamity Relief Fund (CRF) in the same pattern as the Tenth FC recommended, with contributions from the Centre and the States in the ratio of 75:25. The size of the 11th FC was Rs. 11007.59 crore, and its major recommendations and observations include the following:

- For the first time, a paradigm shift from the Response-centric approach to a Risk Reduction approach was felt. Accordingly, the FC recommended to the Central government (ministries), State governments, and Planning Commission to devise medium and long-term measures for reducing, and, if possible, eliminating the occurrence of these calamities by undertaking developmental works.
- As with previous FCs, the Calamity Relief Fund (CRF) was recommended to be kept out of the Public Account of the State and should be invested in a manner approved by the Ministry of Finance.
- The balance of funds at States/UTs, at the end of the 5-year plan, may be made available to states for being used as a resource for the next plan.
- The FC recommended that the Union Ministry of Agriculture continue to be the Nodal Ministry for coordinating relief works.
- The Scheme of the *National Fund for Calamity Relief (NFCR)* was discontinued given the difficulty in unambiguously defining the *Calamity of Rare Nature* and the difficulty in assisting out of the limited resources.
- It was felt that in case of calamity of rarest severity, there is a need to develop a system in which it should be possible to take suo-moto cognizance of the occurrence of calamity. Hence the FC recommended setting up of *"National Calamity Contingency Fund (NCCF)"* with a core Corpus of ₹ 500 crores. It is replenished through the *National Calamity Contingency Duty* imposed on cigarettes, pan masala, beedis, tobacco products and cellular phones.
- Under the NCCF, the Commission recommended establishing a "*National Centre for Calamity Management" (NCCM)* under the Ministry of Agriculture. The aim was to monitor all types of calamities and empower to recommend to the Union Government whether the calamity was severe.
- The FC further recommended that the NCCM should also develop expertise for providing training to the State manpower regularly; undertake monitoring and documentation; and keep an inventory of physical sources available at various places.
- The Commission recommended establishing an interdisciplinary cadre of about 200 to 300 people under the Relief Commissioner in the State, which could be deployed for relief works during and after calamities.

iii) The Twelfth Finance Commission

The recommendations of the Twelfth Finance Commission (**2005-2010**) under the Chairmanship of Dr C Rangarajan marked a shift towards a more structured and accountable approach to disaster management funding in India. By focusing on mitigation and preparedness, the Commission laid the groundwork for a more resilient disaster management system, aligning with the broader goals of sustainable development and risk reduction.

- The system of the Calamity Relief Fund (CRF) was continued as recommended by earlier FCs.
- The size of the CRF was enhanced to Rs. 21333.33 crore, and the Centre: State contribution continued as 75:25.
- The Commission considered that natural calamity, besides floods, drought, cyclones, hailstorms, and fire as applicable in the EFC, also covers landslides, avalanches, cloudbursts and pest attacks.
- The TFC recommended that disaster preparedness and mitigation needs to be built into the State Plan, not in the CRF/NCCF.
- For disaster preparedness, the Commission recommended setting up a Committee of scientists, flood control specialists, and other experts to study the hazards affecting several States/ UTs.
- The NCCF scheme was also recommended to be continued with a core corpus of Rs. 500 crore.
- A new concept of *"Earthquake Pool"* emerged during the TFC. As per IRDA information to the Commission, the General Insurance Company (GIC) has decided to create an earthquake pool where the insurers will divert the earthquake premia to the pool so that the burden of risk arising out of huge losses due to earthquakes may be shared between insurer. This scheme can be very useful for providing social security during any catastrophic event.
- The Centre was recommended to make allocations of food grains as relief material to the needy states but in a transparent manner.

iv) The Thirteenth Finance Commission

The recommendations of the Finance Commission (**2010-15**) under the **Chairmanship** of Dr. Vijay Kelkar were crucial in formalising and strengthening the disaster management framework in India. By introducing dedicated funds for disaster response and mitigation, the Commission recommended creating a more robust financial structure that encouraged both immediate relief efforts and long-term risk reduction strategies. This marked a significant evolution in India's approach to disaster management, aligning it more closely with global best practices. The major recommendations include the following significant points:

- It considered the newly formulated Disaster Management Act, 2005, in its recommendations.
- The Commission initiated two studies to analyse the disaster relief trends and aspects. The first Study focused on the impact of past disasters, the

frequency of disaster occurrences, expenditure on relief between 2005 and 2015, and a review of mitigation schemes undertaken by various ministries and departments at both the Central and State levels. However, the second Study focused on the financing arrangements under the Disaster Management Act, 2005, the effectiveness of the NCCF, the role of finance commissions in financing relief, and the principles of allocation under the CRF.

- The FC recommended that with effect from April 1, 2010, the existing NCCF should be merged with the *National Disaster Response Fund (NDRF)* as per the provisions of the Disaster Management Act, 2005. Any balance in the NCCF at the end of 2009-10 should be transferred to the NDRF. Similarly, the CRF should be merged with the newly created *State Disaster Response Fund (SDRF)* from April 1, 2010, under the DM Act, 2005.
- The recommended total size of the SDRF was ₹ 33,581 crores, to be shared between the Central government and general category States in a 75:25 ratio, and 90:10 for special category States. The Central government's share of 75 per cent and 90 per cent would be classified as *Grants-in-Aid*.
- The administrative mechanism for disaster relief was defined, with the *National Disaster Management Authority (NDMA)* and *National Executive Committee (NEC)* at the Central level, and the *State Disaster Management Authority (SDMA)* and *State Executive Committee (SEC)* at the State level.
- The *Public Liability Insurance (PLI) Act, 1991*, covers over 179 explosives, and toxic and highly reactive chemicals, making businesses involved in hazardous activities legally responsible for any harm caused. *The Environmental Relief Fund (ERF)* was established under the PLI Act in 2008, and by March 31, 2009, the ERF had a total corpus of ₹ 285 crores, strengthening its effectiveness. Although the PLI Act and ERF provide additional financial resources for addressing man-made disasters, the Finance Commission recommended adjusting the relief amounts for inflation and expanding the list of chemicals covered under the Act.
- An additional grant of ₹ 525 crores was recommended for capacity-building in disaster management.

v) The Fourteenth Finance Commission

The recommendations of the FC (**2015-20**) under the **Chairmanship of Dr. Y.V. Reddy** played a crucial role in strengthening the disaster management system in India by ensuring that states had sufficient resources and flexibility to manage disasters effectively. By allowing to use of funds for localised disasters and emphasising preparedness, the Commission's recommendations aimed to build a more resilient and responsive disaster management framework across the country. These recommendations focused on enhancing the financial autonomy and flexibility of states in managing disasters as well as ensuring adequate resources for disaster response and preparedness. Its major recommendations are highlighted below:

- The FC recommended a total corpus of ₹ 1,60,153 crores for states to manage disaster. Out of this, ₹ 1, 22,601 crores were contributed by the Union and ₹ 37,552 crores was contributed by the States/UTs.
- States were allowed to use 10 per cent of the SDRF grants to address localised natural disasters not included in the Ministry of Home Affairs list of notified disasters.
- The Commission recommended giving greater flexibility to the states in utilising disaster management funds, allowing them to effectively roll out disaster response at the local level.

vi) The Fifteenth Finance Commission

The Fifteenth Finance Commission (FFC) under the Chairmanship of Shri N.K. Singh, a former Indian Administrative Services Officer, economist and renowned politician. He has given landmark recommendations in Disaster Risk Management for the fiscal year 2020-21 and 2021-22 to 2025-26. These recommendations have created a new paradigm of Disaster Risk Financing in India. For the first time, the government has shifted focus from the response-centric approach to a disaster risk management approach in a systematic manner. The specific recommendations of the FFC on Disaster Response and Disaster Mitigation per the DM Act, 2005 have set the stage for a paradigm shift to the culture of prevention, preparedness and mitigation.

Unlike the earlier recommendations of previous Finance Commissions, which focused on an expenditure-based approach for determining the allocations of funds for disaster management to the State governments, the FFC has adopted a new methodology based on three specific factors, which provides a more comprehensive framework for allocating funds to states and union territories.

The FFC has submitted two critical reports concerning DRR in India:

i) Report of Fifteenth Finance Commission for 2020-21, and

ii) Report of Fifteenth Finance Commission for 2021-26.

These recommendations, encapsulated in Chapter 6 of the 2020-21 Report and Chapter 8 of the subsequent Report, outline a comprehensive strategy for disaster management, focusing on the allocation of resources, the creation of specialised funds, and a new methodology for fund distribution to states. Traditionally, disaster response funds were available at both the national and state levels through the NDRF and SDRF. However, recognising the importance of not only responding to disasters but also mitigating risks before they occur, the Commission has recommended the creation of broader funds, namely the *National Disaster Risk Management Fund (NDRMF),* and *State Disaster Risk Management Fund (SDRMF)* at the national level and state level respectively. The NDRMF is further categorised into two parts – the *National Disaster Response Fund* (NDRF) and the *National Disaster Mitigation Fund (NDMF).* Similarly, the SDRMF is categorised into the *State Disaster Response Fund(SDRF)* and the *State Disaster Mitigation Fund (SDMF)*.

The 15th FC also recommended a revised contribution structure for state allocations, wherein the general contribution rate is set at 25 per cent, for all states/UTs except for the North-Eastern and Himalayan (NEH) states, which contribute only 10 per cent. This adjustment acknowledges the unique vulnerabilities, and financial constraints of the NEH states while ensuring adequate disaster preparedness and response capabilities.

New Methodology of Fund Allocation

The FFC has introduced a new methodology for the allocation of disaster management funds, deviating from the expenditure-based approach of previous Commissions. This new methodology is a hybrid model combining the following three key factors:

- ***Capacity-*** reflected through past expenditure on disaster management
- ***Risk Exposure***- determined by the area and population of the states
- ***Hazard & Vulnerability***- assessed through a disaster risk index, incorporating factors such as geographical and socio-economic vulnerabilities.

This approach ensures a more balanced and need-based allocation of resources to the states, enhancing the effectiveness of disaster management efforts. Out of the total grants earmarked for disaster management, 20 per cent is allocated to mitigation efforts, while the remaining 80 per cent is designated for response-related activities. The response fund itself is further divided into three windows as mentioned below:

i) ***Response and Relief–*** 40 per cent of the total NDRMF/SDRMF is allocated to immediate disaster response and relief operations.

ii) ***Recovery and Reconstruction*** –30 per cent of the total is dedicated to rebuilding assets and livelihoods after disasters, addressing the gap in resources traditionally allocated to recovery efforts.

iii) ***Preparedness and Capacity-Building***–10 per cent of the total allocated NDRMF/SDRMF aimed at enhancing state-level preparedness, supporting State Disaster Management Authorities (SDMAs), training, capacity-building activities, and emergency response infrastructure.

Box 1: Fund Allocation at a Glance

1) **National Disaster Risk Management Fund -** A total corpus of **₹68,463**croress has been allocated for the period 2021-2026 which is further allocated as under:
   - ***National Disaster Response Fund*** - ₹54,770 crores (80% of NDRMF)
   - ***National Disaster Mitigation Fund-*** ₹13,693 crores (20% of NDRMF)
2) **State Disaster Risk Management Fund** - A total of **₹1,60,153**croress has been allocated for the period 2021-2026 which is further allocated as under:
   - ***State Disaster Response Fund–*** ₹ 1,28,122 crores (80% of SDRMF)

National Disaster Risk Management Fund

First time in the history of FC recommendations, much emphasis has been given to Disaster Risk Management in the country. At the national level, a specific *National Disaster Risk Management Fund (NDRMF)* has been recommended by the Fifteenth FC. This is further divided into two sub-heads–*National Disaster Response Fund (NDRF)* and *National Disaster Mitigation Fund (NDMF).*

Separate allocations have been made for both the NDRF and the NDMF to address various critical areas. The NDRF has allocated Rs. 5,000 cr. for the expansion and modernisation of fire services and ₹1,000 crores for the resettlement of people displaced by erosion. From the NDMF, ₹1,200 crores have been designated for catalytic assistance to the twelve most drought-prone states, while ₹750 crores are allocated for managing seismic and landslide risks in ten hill states. Additionally, ₹2,500 crores is earmarked for reducing the risk of urban flooding in seven of the most populous cities, and ₹1,500 crores is set aside for mitigation measures to prevent erosion.

The FFC has also recommended a cost-sharing mechanism for the central assistance provided through the NDRF and NDMF. This arrangement is designed to encourage states to contribute to disaster management while avoiding excessive demands on central resources. The recommended cost-sharing structure is as follows:

i) ***10 per cent State Contribution*** for the central assistance up to ₹250 crores

ii) ***20 per cent State Contribution f***or assistance up to ₹500 crores

iii) ***25 per cent State Contribution*** for assistance exceeding ₹500 crores.

State Disaster Risk Management Fund

The State Disaster Risk Management Fund (SDRMF) has been allocated a total corpus of ₹1, 60,153crores for the period 2021-26, with the Union government contributing ₹1,22,601crores and the states adding ₹37,552 crores. Like the NDRMF, the SDRMF is also divided into the State Disaster Response Fund (SDRF), which receives 80 per cent, and the State Disaster Mitigation Fund (SDMF), which receives 20 per cent of the total allocation. The provisions within the SDRMF are designated for specific purposes, including ₹64,061 crores (40%) for Response and Relief, ₹48,046 crores (30%) for Recovery and Reconstruction, and ₹16,015 crores (10%) for Preparedness and Capacity-Building.

Now we will depict the allocation and release of funds from the SDRMF/ SDRF and NDRF during 2020-2021 – as of 31.03.2021 in the following table (8.1).

Table 8.1: Allocation and Release of Funds from SDRMF/ SDRF and NDRF during 2020-2021

As on 31st March 2021

(Rs. in crore)

Sl. No.	Name of the State	Allocation of SDRMF			Releases from SDRMF		Releases from NDRF
		Central Share	State Share	Total	1st Instalment	2nd Instalment	
1	2	3	4	5	6	7	8
1.	Andhra Pradesh	1,119.00	372.00	1,491.00	559.50	559.50	657.029
2.	Arunachal Pradesh	250.00	28.00	278.00	125.00	125.00	59.34
3.	Assam	772.00	86.00	858.00	386.00	386.00	44.37
4.	Bihar	1,416.00	472.00	1,888.00	708.00	708.00	1255.27
5.	Chattisgarh	432.00	144.00	576.00	216.00	216.00	--
6.	Goa	12.00	3.00	15.00	6.00	6.00	--
7.	Gujarat	1,324	441.00	1,765.00	662.00	662.00	--
8.	Haryana	491.00	164.00	655.00	245.50	245.50	--
9.	Himachal Pradesh	409.00	45.00	454.00	204.50	204.50	2.90
10.	Jharkhand	568.00	189.00	757.00	284.00	284.00	--
11.	Karnataka	791.00	263.00	1.054.00	395.50	395.50	689.27
12.	Kerala	314.00	105.00	419.00	157.00	157.00	--
13.	Madhya Pradesh	1,820	607.00	2,427.00	910.00	910.00	1891.79
14.	Maharashtra	3,222	1074.00	4,296.00	1,611	1,611	420.12

15.	Manipur	42.00	5.00	47.00	21.00	21.00	26.53
16.	Meghalaya	66.00	7.00	73.00	33.00	33.00	16.52
17.	Mizoram	47.00	5.00	52.00	23.50	23.50	--
18.	Nagaland	41.00	5.00	46.00	20.50	20.50	1.335
19.	Odisha	1,604.00	535.00	2,139.00	802.00	802.00	500.00
20.	Punjab	495.00	165.00	660.00	325.925	247.50	--
21.	Rajasthan	1,481.00	494.00	1,975.00	740.50	740.50	68.65
22.	Sikkim	50.00	6.00	56.00	25.00	25.00	73.86
23.	Tamil Nadu	1,020.00	340.00	1,360.00	510.00	510.00	286.91
24.	Telangana	449.00	150.00	599.00	224.50	224.50	--
25.	Tripura	68.00	8.00	76.00	34.00	34.00	12.93
26.	Uttar Pradesh	1,933.00	645.00	2,578.00	966.00	966.00	--
27.	Uttarakhand	937.00	104.00	1,041.00	468.50	468.50	--
28.	West Bengal	1,011.00	337.00	1,348.00	505.50	505.50	2250.28
	Total	**22,184.00**	**6,799.00**	**28,983.00**	**11.170.425**	**11,092.00**	**8,257.11**

Source: MHA, Government of India

Note: Installments of the centre's share of SDMF/SDRMF are released to the State Government, subject to the submission of the Utilisation Certificate. Annual report, etc. Budgetary provisions and releases of the SDRMF/SDRF and NDRF are dealt with by the Department of Expenditure.

Hence, the Grant-in-aid of **Rs. 30,519.535 crore was** released during the financial year **2020-2021.**

The data presented in the following table (8.2), depicts the state-wise allocations of SDRF.

Table 8.2: State-wise allocation of State Disaster Response Fund during 2021-2026
(including Central as well as State share)

(Rs. in crore)

Sl. No.	**Name of the State**	**2021-22**	**2022-23**	**2023-24**	**2024-25**	**2025-26**	**Total 2021-26**
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
1.	Andhra Pradesh	1192.80	1252.80	1315.20	1380.80	1449.60	6591.20
2.	Arunachal Pradesh	222.40	233.60	245.60	256.80	270.40	1228.80
3.	Assam	686.40	720.80	756.80	795.20	834.40	3793.60
4.	Bihar	1510.40	1586.	1664.80	1748.00	1836.00	8345.60

			40				
5.	Chattisgarh	460.80	484.00	508.00	533.60	560.00	2546.40
6.	Goa	12.00	12.80	12.80	13.60	15.20	66.40
7.	Gujarat	1412.00	1482.40	1556.80	1635.20	760.00	7802.40
8.	Haryana	524.00	550.40	577.60	606.40	636.80	2895.20
9.	Himachal Pradesh	363.20	380.80	400.80	420.00	441.60	2006.40
10.	Jharkhand	605.60	635.20	667.20	701.60	736.00	3345.60
11.	Karnataka	843.20	885.60	929.60	976.00	1024.80	4659.20
12.	Kerala	335.20	352.00	369.60	388.00	408.00	1852.80
13.	Madhya Pradesh	1941.60	2038.40	2140.80	2248.00	2360.00	10728.80
14.	Maharashtra	3436.80	3608.80	3788.810	3978.40	4176.80	18989.60
15.	Manipur	37.60	39.20	41.60	44.00	45.00	208.00
16.	Meghalaya	58.40	60.80	64.80	67.20	71.20	322.40
17.	Mizoram	41.60	43.20	46.40	48.00	50.40	229.60
18.	Nagaland	36.80	38.40	40.80	42.40	44.80	203.20
19.	Odisha	1711.20	1796.80	1886.40	1980.80	2080.00	9455.20
20.	Punjab	528.00	554.40	582.40	611.20	642.40	2918.40
21.	Rajasthan	1580.00	1659.20	1742.40	1828.80	1920.00	8730.40
22.	Sikkim	44.80	47.20	49.60	52.00	54.40	248.00
23.	Tamil Nadu	1088.00	1142.40	1200.00	1260.00	1322.40	6012.80
24.	Telangana	479.20	503.20	528.00	555.20	582.40	2648.00
25.	Tripura	60.80	63.20	67.20	70.40	74.40	336.00
26.	Uttar Pradesh	2062.40	2165.60	2273.60	2388.00	2507.20	11396.80
27.	Uttarakhand	832.80	874.40	918.40	964.00	1012.00	4601.60
28.	West Bengal	1078.40	1132.80	1189.60	1248.00	1311.20	5960.00
	Total	**23186.40**	**24344.80**	**25565.60**	**26841.60**	**28184.00**	**128122.40**

Source: MHA, Government of India

The SDRF has been established under Section 48 (1) (a) of the Disaster Management Act, 2005. It is a vital financial resource allocated to the state/UT government in its disaster response efforts. This fund is specifically allocated for providing immediate relief to the affected communities due to *officially notified* *disasters*, focusing solely on the immediate aftermath rather than on long-term recovery or reconstruction.

A significant portion of the SDRF is funded by the Central Government, which contributes 75 per cent of the allocation for general category States/UTs and 90 per cent for special category States/UTs including the North Eastern States, Sikkim, Uttarakhand, Himachal Pradesh, and Jammu & Kashmir. This central contribution is distributed in two equal installments each year, based on the recommendations of the Finance Commission.

The SDRF is designed to respond to a wide array of natural disasters, such as cyclones, droughts, earthquakes, fires, floods, tsunamis, hailstorms, landslides, avalanches, cloudbursts, pest attacks, frost, and cold waves. In addition, states are granted some flexibility to address local disasters that may not be included in the national list of notified disasters up to 10 per cent of the SDRF can be used by a State government to provide relief for such local disasters, provided they are recognised as significant within the state's context, even if not officially listed by the Ministry of Home Affairs. It requires a clear, transparent, and approved set of norms and guidelines for responding to these local disasters, with approval from the State SEC.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Explain the evolution of disaster management funding in India as recommended by various Finance Commissions.

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2) "The FFC has introduced a new methodology for the allocation of disaster management funds". Discuss.

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8.4 DISASTER RISK FINANCING, INSURANCE AND RISK TRANSFER

Disaster Risk Finance, Insurance, and Risk Transfer are essential financial strategies for managing the economic impacts of disasters. These mechanisms provide financial protection and resilience against the economic losses caused by natural hazards.

Disaster Risk Finance involves the planning and allocation of financial resources before a disaster strikes, ensuring that funds are readily available to support relief, recovery, and reconstruction efforts. In many countries, Insurance is being used as a proactive tool for post-disaster recovery. Risk transfer through insurance and reinsurance solutions has played a crucial role in supporting government efforts to close the protection gap. It plays an important role wherein the financial burden of potential losses from individuals, businesses, and governments is transferred to insurance companies, thereby spreading risk across a broader base. Risk transfer mechanisms, such as *Parametric Insurance* and *Catastrophe Bonds* further enhance financial resilience by providing swift pay-outs based on predefined triggers, such as the occurrence of a specific disaster or natural calamity. Together, these tools enable governments, communities, and businesses to reduce their vulnerability to disasters, ensuring that they can respond quickly and effectively, minimising economic disruption, and accelerating recovery efforts.

The Government of India has introduced several social protection schemes utilising insurance solutions, such as Ayushman Bharat Pradhan Mantri Jan Arogya Yojana, Pradhan Mantri Jeevan Jyoti Bima Yojana, Pradhan Mantri Suraksha Bima Yojana, Pradhan Mantri Fasal BimaYojana, and Atal Pension Yojana. For such a scheme to be effective, timely assistance must be ensured to beneficiaries in their moments of need, especially during the event of calamity or disaster.

First of all, we will highlight the need and significance of the Pradhan Mantri Fasal Bima Yojana (PMFBY) in Disaster Management.

i) **Pradhan Mantri Fasal Bima Yojana**

The PMFBY is a Government of India's initiative, aimed at providing comprehensive insurance coverage to farmers against crop losses due to unforeseen disasters or calamities. It was primarily launched to support sustainable agricultural practices across the country. Under this Scheme, financial assistance is given to farmers in case of crop damage due to disasters thereby, helping them to stabilise their income and ensure their continued participation in agricultural practices.

Weather data plays a crucial role in determining the extent of damage due to calamity at a particular place. The scheme also encourages the adoption of innovative and modern farming techniques, and it plays a crucial role in facilitating the flow of credit to the agricultural sector. By mitigating production risks, the PMFBY contributes to food security, promotes crop diversification, and enhances the growth and competitiveness of agriculture in India.

ii) **Parametric Insurance**

It is one of the innovative and effective examples of Disaster Risk Financing that provides pay-outs based on the occurrence of disasters/hazards as per the pre-defined parameters, rather than on the assessment of actual losses.

In traditional insurance, a detailed evaluation of damage and loss is required after an event, for which historical data is required to be available as evidence whereas, in parametric insurance, pay-outs are activated immediately when specific conditions are met such as a certain level of rainfall, wind speed, or earthquake magnitude. This approach allows for faster, more transparent, and more predictable disbursements, making it particularly useful in disaster-prone areas where quick financial relief is crucial. By relying on objective data and predefined parameters, parametric insurance contributes to reducing administrative costs and speeds up the recovery process, offering a valuable financial safety net for individuals, businesses, and governments.

Here are a few instances of parametric insurance being utilised in India:

a) The PMFBY verifies crop losses, while the Weather-based Crop Insurance Scheme (WBCIS) employs threshold limits to determine the payouts.

b) In Kerala, the Co-operative Milk Marketing Federation offers parametric insurance to safeguard dairy farmers against the effects of heat stress.

Nagaland is the first Indian state to adopt parametric insurance for extreme precipitation, with the policy tailored to weather data specific to certain locations.

8.5 FINANCIAL ARRANGEMENTS AND DISASTER MANAGEMENT: WAY FORWARD

The 15th Finance Commission's recommendations on Disaster Risk Management and Disaster Response have significantly changed how we approach disaster resilience. Unlike the 10th Finance Commission, which mainly focuses on calamity relief, the 15th Finance Commission emphasises both disaster response and comprehensive resilience planning, including financial allocations.

Effective financial management is very important to ensure that the resources are available at the right place, at the right time, and to the right stakeholders when needed. This is vital for strengthening resilience at all levels and in the communities.

The need of the hour is to adopt proactive, inclusive, and transparent approaches to manage financial risks from disasters and support sustainable development goals. Now, it is the time for policy planners to focus on disaster resilience at the grassroots level with increased financial allocations. More funding should be directed towards multi-hazard forecasting, early warning systems, last-mile connectivity, and structured capacity-building interventions.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Explain the role of insurance and risk transfer mechanisms in disaster risk financing.

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2) What are the necessary measures to ensure effective financial arrangements for disaster management?

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8.6 CONCLUSION

This Unit has provided a detailed overview of the financial mechanisms and management strategies for disaster management in India, emphasising the evolution of funding arrangements through recommendations of the various Finance Commissions. It focused on the integration of disaster risk financing within the broader framework of disaster management, highlighting the importance of insurance and risk transfer mechanisms in mitigating financial losses. The discussion in this Unit, underscored the need for a proactive approach, focusing on preparedness and resilience while ensuring that the financial resources are effectively allocated and utilised to enhance the capacity for disaster risk reduction and management.

8.7 GLOSSARY

Calamity of Rare Nature : It refers to a catastrophic event that occurs infrequently but has severe consequences and which is beyond the coping capacity of local or regional authorities. For example, extreme disasters like massive earthquakes, tsunamis, or pandemics are rare in occurrence but have a significant impact on lives, infrastructure, and economies when they occur.

Disaster Risk Finance : It includes financial strategies and instruments designed for managing the economic and fiscal impacts of disasters. This includes mechanisms like insurance. In this context, the Pradhan Mantri Fasal Bima Yojana, Weather-based Crop Insurance Scheme, Catastrophic Bond, etc. ensure that sufficient financial resources are available to respond to and recover from disasters. The goal is to reduce the financial vulnerability of governments, businesses, and communities to natural and man-made hazards.

Financial Arrangement : In the context of disaster risk management, the term Financial Arrangement refers to the set of financial instruments and strategies put in place to manage the risks associated with disasters. This can be the National Disaster Risk Management Fund, State Disaster Risk Management Fund, Contingency Fund, ex-Gratia, etc.

Insurance : It is a financial product that protects against specific risks by transferring the financial burden from an individual or organisation to an insurance company. In disaster risk management, insurance helps to cover losses caused by events such as natural disasters, accidents, or other unforeseen calamities. The insured pays a premium, and in return, the insurer compensates for losses covered under the policy.

Risk Transfer : It is the process of shifting the financial burden of a risk to another party, usually through mechanisms such as insurance or reinsurance. Instead of retaining the risk and its associated costs, an organisation or individual transfers it to an insurer, who assumes the financial responsibility in exchange for a premium. This is a key strategy in managing the financial impacts of disasters and reducing the exposure to potentially devastating losses.

Ex-Ante Investment : This refers to the financial resources allocated and actions taken before a disaster or an adverse event occurs. These investments are proactive and aimed at reducing the potential impacts of future risks. For example, funding for infrastructure resilience, early warning systems, disaster preparedness training, risk assessment, etc.

Ex-Post Investment : It is defined as the financial resources and actions taken after a disaster or an adverse event has occurred. These investments are reactive and focus on recovery, reconstruction, and rehabilitation. For example, rebuilding damaged infrastructure, providing emergency relief, and compensating affected communities. Hence, the ex-post investment aims to restore normalcy and address the immediate and long-term impacts of the disaster.

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8.9 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Your answer should include the following points:
   - For details, refer to Section 8.3
2) Your answer should include the following points:

   The 15th FC has introduced a new methodology for the allocation of disaster management funds, deviating from the expenditure-based approach of previous Commissions. This new methodology is a hybrid model combining the following three key factors:

   a) ***Capacity-*** *reflected through past expenditure on disaster management.*

   b) ***Risk Exposure****- determined by the area and population of the states.*

   c) ***Hazard & Vulnerability****- assessed through a disaster risk index, incorporating factors such as geographical and socio-economic vulnerabilities.*

   - For more details, refer to Section 8.3 (vi)

Check Your Progress 2

1) Your answer should include the following points:
   - For details, refer to Section 8.4
2) Your answer should include the following points:
   - For details, refer to Section 8.5